

The generalization of Theorem 3(d) to products of more than two factors can be stated in words as follows:

The transpose of a product of matrices equals the product of their transposes in the reverse order.

Practice Problems

1. Since vectors in $\mathbb{R}^n$ may be regarded as $n \times 1$ matrices, the properties of transposes in Theorem 3 apply to vectors, too. Let

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \text{ and } x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

Compute $(Ax)^T$, $x^T A^T$, xx^T and $x^T x$. Is $A^T x^T$ defined?

Solution:

$$Ax = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + (-3) \cdot 3 \\ (-2) \cdot 5 + 4 \cdot 3 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$x^T = \begin{bmatrix} 5 & 3 \end{bmatrix} \quad (\text{By defn. of transpose}) \quad A^T = \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} \quad (\text{By defn. of transpose})$$

$$x^T A^T = \begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 5 \cdot 1 + 3 \cdot (-3) & 5 \cdot (-2) + 3 \cdot 4 \end{bmatrix} = \begin{bmatrix} -4 & 2 \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$xx^T = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \begin{bmatrix} 5 & 3 \end{bmatrix} = \begin{bmatrix} 5 \times 5 & 5 \times 3 \\ 3 \times 5 & 3 \times 3 \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$x^T x = \begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \times 5 + 3 \times 3 \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$= \begin{bmatrix} 25 & 15 \\ 15 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 36 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} \quad x^T = \begin{bmatrix} 5 & 3 \end{bmatrix}. \quad (\text{From the definition of transpose})$$

A^T is a 2×2 matrix and x^T is a 1×2 matrix. By defn. of matrix multiplication, $A^T x^T$ is not defined since A^T has 2 columns which is not the same as 1 row of x^T .